

BEE 208 — Question 20: Skip Negative Power Readings

Control Statement: for loop with continue | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables reading, totalPower (double) and i (int)

Step 3: Set totalPower = 0

Step 4: Use a for loop from i = 1 to 10

Step 5: Prompt user to enter power reading i

Step 6: Read reading

Step 7: If reading < 0, display 'Invalid reading skipped.' and continue

Step 8: totalPower = totalPower + reading

Step 9: End loop

Step 10: Display totalPower of valid readings

Step 11: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE reading, totalPower AS DOUBLE  
  
DECLARE i AS INTEGER  
  
SET totalPower = 0  
  
FOR i = 1 TO 10 DO  
    OUTPUT 'Power Reading', i, '(W):'  
  
    INPUT reading  
  
    IF reading < 0 THEN  
        OUTPUT 'Invalid reading skipped.'  
        CONTINUE  
    END IF  
  
    totalPower = totalPower + reading  
  
END FOR  
  
OUTPUT 'Total Valid Power:', totalPower, 'W'  
  
END
```

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	Readings: 100,-5,200,300,-10,400,500,600,700,800	Expected Output	Total = 3600 W
Input	10 valid positive readings	Expected Output	Sum of all 10 readings